

Problem Statement and Goals

Software Engineering

Team 13, Keystone

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Table 1: Revision History

Date	Developer(s)	Description of Changes
Sept 25th 2026	Ayush Patel	Added custom documents
Sept 26th 2026	Waqar Ul-Hassan	Added Goals and Stretch Goals
Sept 26th 2026	Waqar Ul-Hassan	Added reflection information
Sept 27th 2026	Waqar Ul-Hassan	Updated Goals and Stretch Goals based on feedback
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1 Problem Statement

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1.1 Problem

1.2 Inputs and Outputs

1.3 Stakeholders

1.4 Environment

2 Goals

The primary goal of the platform is to provide a centralized and structured system for managing the capstone project process. The platform will improve the

experience for students, academic supervisors, industry sponsors, and administrators by making project information easier to access, manage, and coordinate.

2.1 Core Goals

1. **Centralized Capstone Project Management:** Provide a single platform where capstone projects can be submitted, viewed, managed, and tracked throughout their lifecycle.
2. **Project Discovery:** Allow students to search and filter available projects based on factors such as skills, technologies, project type, industry, supervisor, and availability.
3. **Student and Stakeholder Profiles:** Allow students, academic supervisors, and industry sponsors to create profiles containing relevant information such as skills, interests, experience, and project preferences.
4. **Team Formation:** Allow students to indicate their intent to form or join a team, and provide visibility into other students seeking teammates for the same or similar projects.
5. **Project Interest and Application Management:** Provide students with the ability to express interest in projects and allow supervisors as well as sponsors to review and manage interested students.
6. **Standardized Project Submissions:** Provide a consistent project submission format that ensures important information, such as objectives, requirements, required skills, and expected deliverables, is clearly presented.
7. **Administrative Management:** Provide administrators with tools to manage users, projects, project availability, and project assignments.

2.2 Stretch Goals

The following features may be implemented if sufficient development time and resources are available.

1. **Team Compatibility Matching:** Recommend potential teammates with respect to their skills, interests, and project preferences.
2. **Intelligent Project Recommendations:** Create a recommendation system that makes suggestions and evaluates compatibility between projects and students based on the student's skills, interests, and preferences.
3. **Analytics Dashboard:** Provide administrators with statistics regarding project demand, student engagement, applications, and project availability.

3 Custom Documents (CDs)

- **User/Stakeholder/Client Interview Report** – Documents the interviews conducted with users, stakeholders, and clients to gather requirements for the application.
- **Usability Testing Report** – Documents the usability testing of the application, including the goals of the testing, the specific tasks we ask users to complete, the specific questions we ask users, and the results

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

Waqar Ul-Hassan:

1. What went well while writing this deliverable?

The team has been awesome in picking up work and I think that's been the best part of it so far. We haven't had any issues with assigning work and infact most of us have been trying to get more work to make sure that we are all keeping up. Furthermore, being able to get to know my team members are bit better as I didn't know two of them before this project has been nice as well. We also had to drop our original project we had committed to and I think our team recovered from that pretty well, I was a bit worried initially but we got everything together quite quickly!

2. What pain points did you experience during this deliverable, and how did you resolve them?

I had a couple pain points through this process, the first was of course the fact that our first project ended up being somewhat complete already which was something that caught us off-guard. I Feel as though we did a great job in resolving this by communicating with our supervising professor as well as Dr. Smith and then ensuring we had a backup project incase we wanted to switch. Another Pain point I had was when deciding some of the metrics for how to track all the work being done equally. The solution to this was just holding discussions, we discussed metrics like commits, number of issues, and lines of code. Overall, we ended up deciding on the number assigned issues as the metric because we felt we could best distribute that one.

3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

We initially had some more particular problem solutions for our project which we faced personally in finding a supervisor, projects, and navigating through all the information in general. After the discussion with our TA

Chris I realized that maybe we needed one or two more smaller things that we should aim to reach as part of our main goal. With this in mind we had discussions with each other and decided on what our goals and stretch goals are now. I also realize as we meet with stakeholders these requirements could change and our TA had said that in the case that they do, we can discuss it with him to update our goals to ensure everything makes sense.